

Problem 25.20

Which of the following are true?

- (I) $\left(\bar{a}_{\overline{n}|} - \frac{d}{\delta}\right)(1+i) = \bar{a}_{\overline{n-1}|}$.
- (II) The present value of a 10 year annuity immediate paying 10 per month for the first eight months of each year is $120a_{\overline{10}|}a_{\frac{(12)}{8/12}}$.
- (III) The present value of a perpetuity paying one at the end of each year, except paying nothing every fourth year, is $\frac{s_{\overline{3}|}}{is_{\overline{4}|}}$.

Solution

Statement I.

We know that

$$\bar{a}_{\overline{n}|} = \frac{1-v^n}{\delta}.$$

Then

$$\left(\bar{a}_{\overline{n}|} - \frac{d}{\delta}\right)(1+i) = \left(\frac{1-v^n}{\delta} - \frac{d}{\delta}\right)(1+i).$$

Since

$$d(1+i) = i,$$

we get

$$\left(\frac{1-v^n-d}{\delta}\right)(1+i) = \frac{1+i-v^{n-1}-i}{\delta}.$$

Thus,

$$\left(\bar{a}_{\overline{n}|} - \frac{d}{\delta}\right)(1+i) = \frac{1-v^{n-1}}{\delta} = \bar{a}_{\overline{n-1}|}.$$

Therefore, statement I is true.

Statement II.

The payments are 10 per month for the first eight months of each year. Since the payment rate is 10 per month, this is equivalent to 120 per year payable monthly. The present value of the first eight monthly payments in each year is

$$120a_{\frac{(12)}{8/12}}.$$

This pattern occurs at the beginning of each of the 10 years, so the annual factor should be an annuity-due factor:

$$\ddot{a}_{\overline{10}|}.$$

Therefore, the present value should be

$$120a_{\frac{8}{12}|}^{(12)}\ddot{a}_{\overline{10}|},$$

not

$$120a_{\overline{10}|}a_{\frac{8}{12}|}^{(12)}.$$

Therefore, statement II is false.

Statement III.

The perpetuity pays at times

$$1, 2, 3, 5, 6, 7, 9, 10, 11, \dots$$

and skips every fourth year. Its present value is

$$v + v^2 + v^3 + v^5 + v^6 + v^7 + \dots$$

Factor this as

$$(v + v^2 + v^3)(1 + v^4 + v^8 + \dots).$$

Thus,

$$PV = (v + v^2 + v^3) \frac{1}{1 - v^4} = \frac{v + v^2 + v^3}{1 - v^4}.$$

Multiplying numerator and denominator by $(1 + i)^4$ gives

$$PV = \frac{(1 + i)^3 + (1 + i)^2 + (1 + i)}{(1 + i)^4 - 1}.$$

However,

$$\frac{s_{\overline{3}|}}{is_{\overline{4}|}} = \frac{(1 + i)^3 - 1}{i} \cdot \frac{1}{(1 + i)^4 - 1} = \frac{1 + (1 + i) + (1 + i)^2}{(1 + i)^4 - 1}.$$

This is not equal to the present value above. Therefore, statement III is false.

Hence, only statement I is true:

$$\boxed{\text{I only}}.$$